



Novel Computational and Machine Learning Approaches Shed Light on Parkinson's Disease Heterogeneity and Subtypes

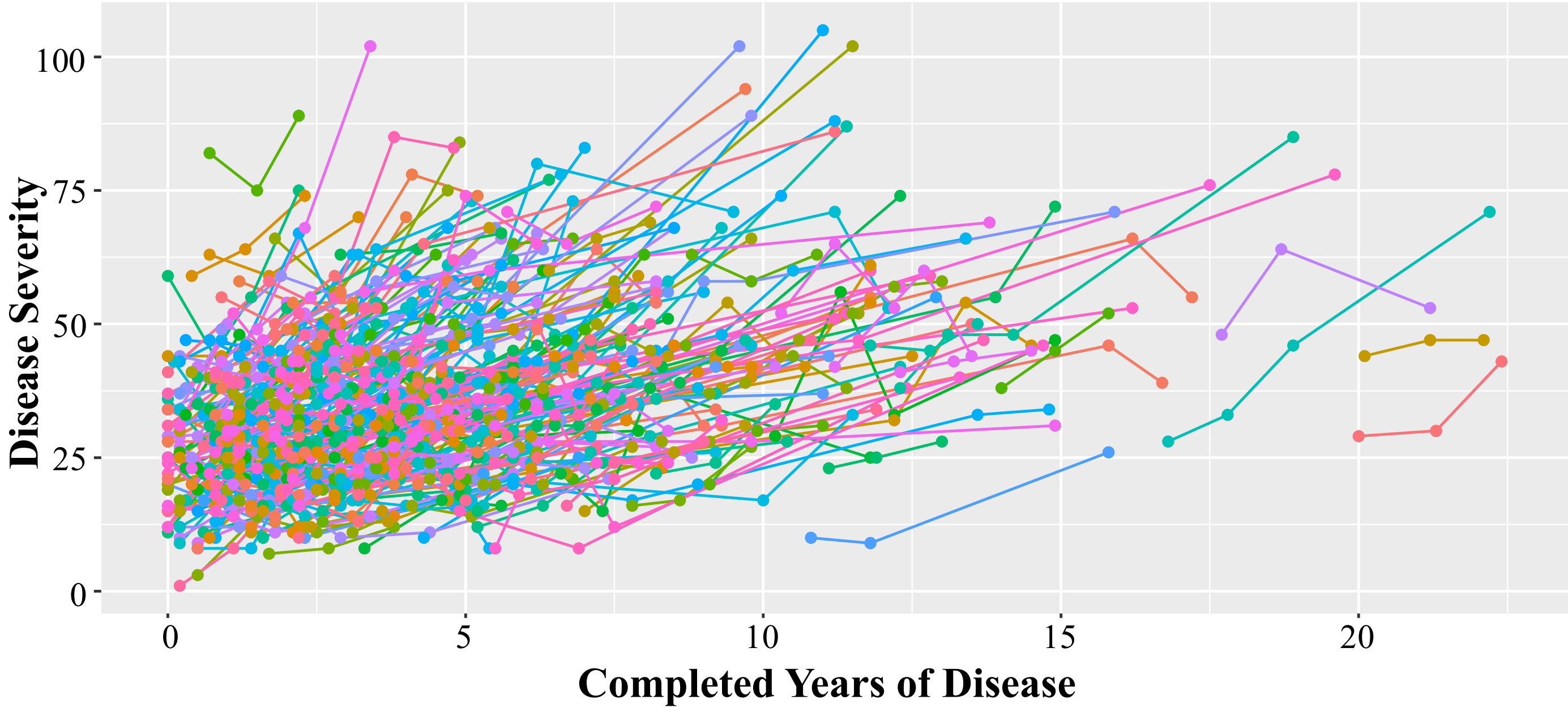
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1. Background

- PD is a **neurodegenerative disease** characterized by formation of **Lewy Body structures** in the brain that degrade dopaminergic neurons
- PD is classified as one of the most **heterogenous diseases** making it difficult to classify as a **unified disorder**



- Patient data from the **Harvard Biomarker Study** were used for this data analysis **modeling patients**, creating **subtypes** and implementing a **predictive model**
- Crucial for **discovering biomarkers** and creation of **personalized clinical trials**

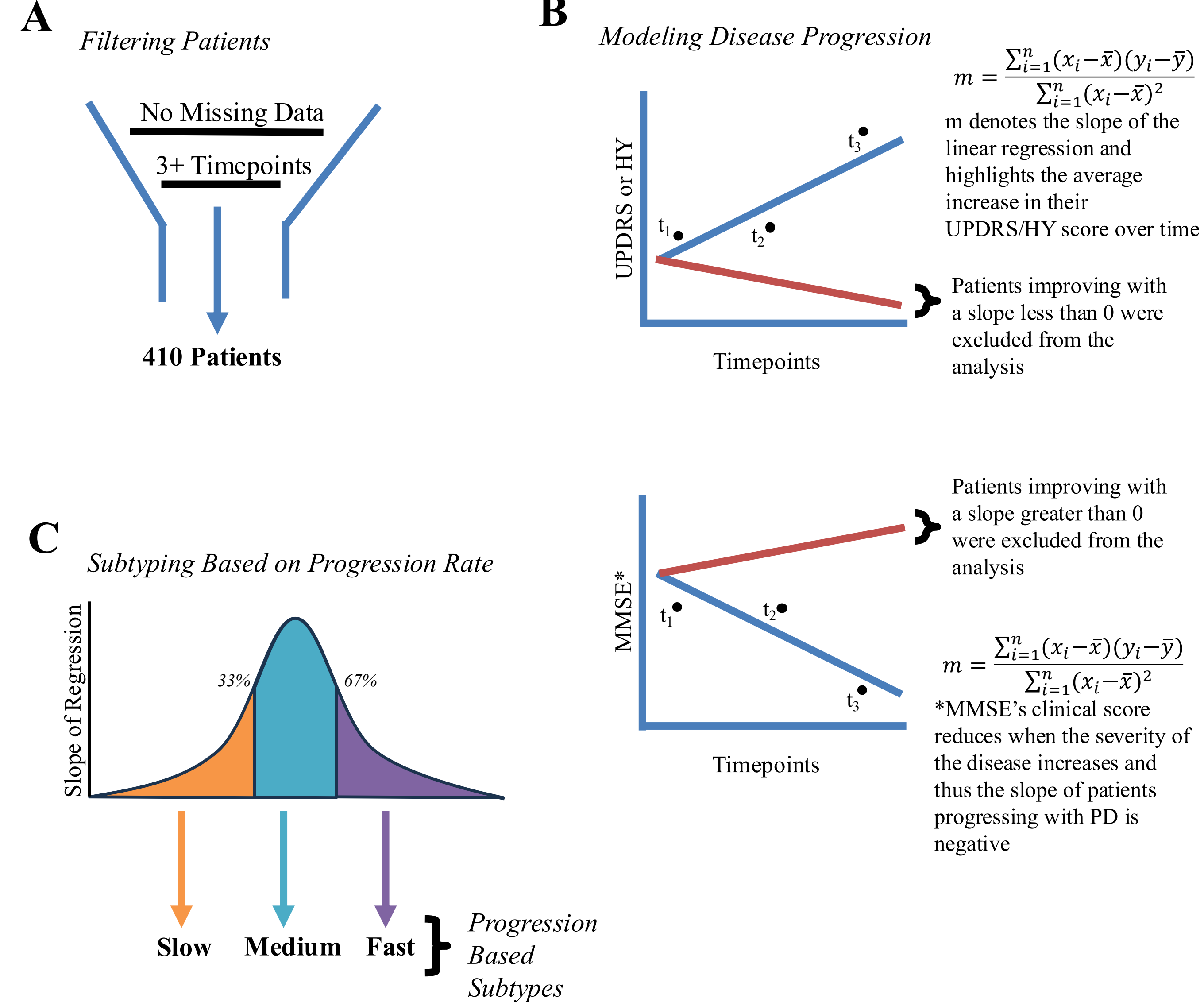
2. Study Population

	UPDRS	HY	MMSE
	N = 318	N = 309	N = 236
Age, years [mean ± SD]	65.7 ± 9.3	65.1 ± 10.0	65.6 ± 9
Sex, female [n (%)]	33.70%	35.70%	32.20%
Race, white [n (%)]	97.50%	97.40%	98.70%
Baseline Disease Duration, years [mean ± SD]	2.85 ± 3.24	2.87 ± 3.25	2.70 ± 2.88

- MMSE (Mini-Mental State Examination):** Assesses cognitive symptoms
- HY (Hoehn & Yahr Scale):** Measures motor symptom severity
- UPDRS (Unified Parkinson's Disease Rating Scale):** Evaluates both motor and cognitive symptoms of PD
- Disease Duration** was calculated from the years since PD diagnosis

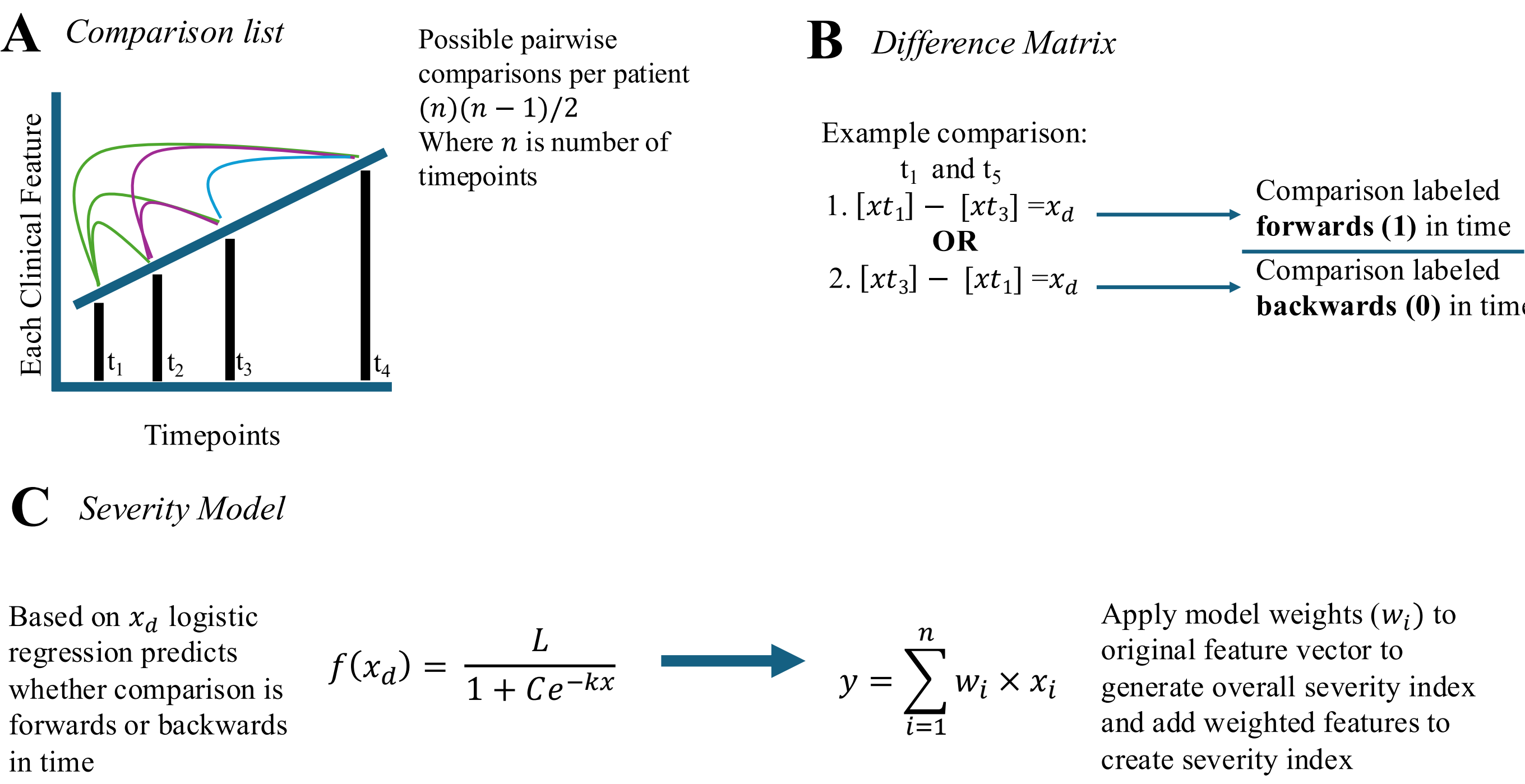
3. Methods and Materials

Clinical Modeling of Patients and Creation of Subtypes



- Patients were **filtered** for having the most data available
- Progression modeling involved extracting the **slope of a linear regression** fit to **clinical scores over total disease duration** (rooted in diagnosis of PD)
- Because PD severity** was assessed using three distinct clinical scales, this led to the identification of **three cohorts**.
- Patients were classified as **fast, medium, or slow** subtypes based on the **top, middle, and lowest terciles** of slope values.

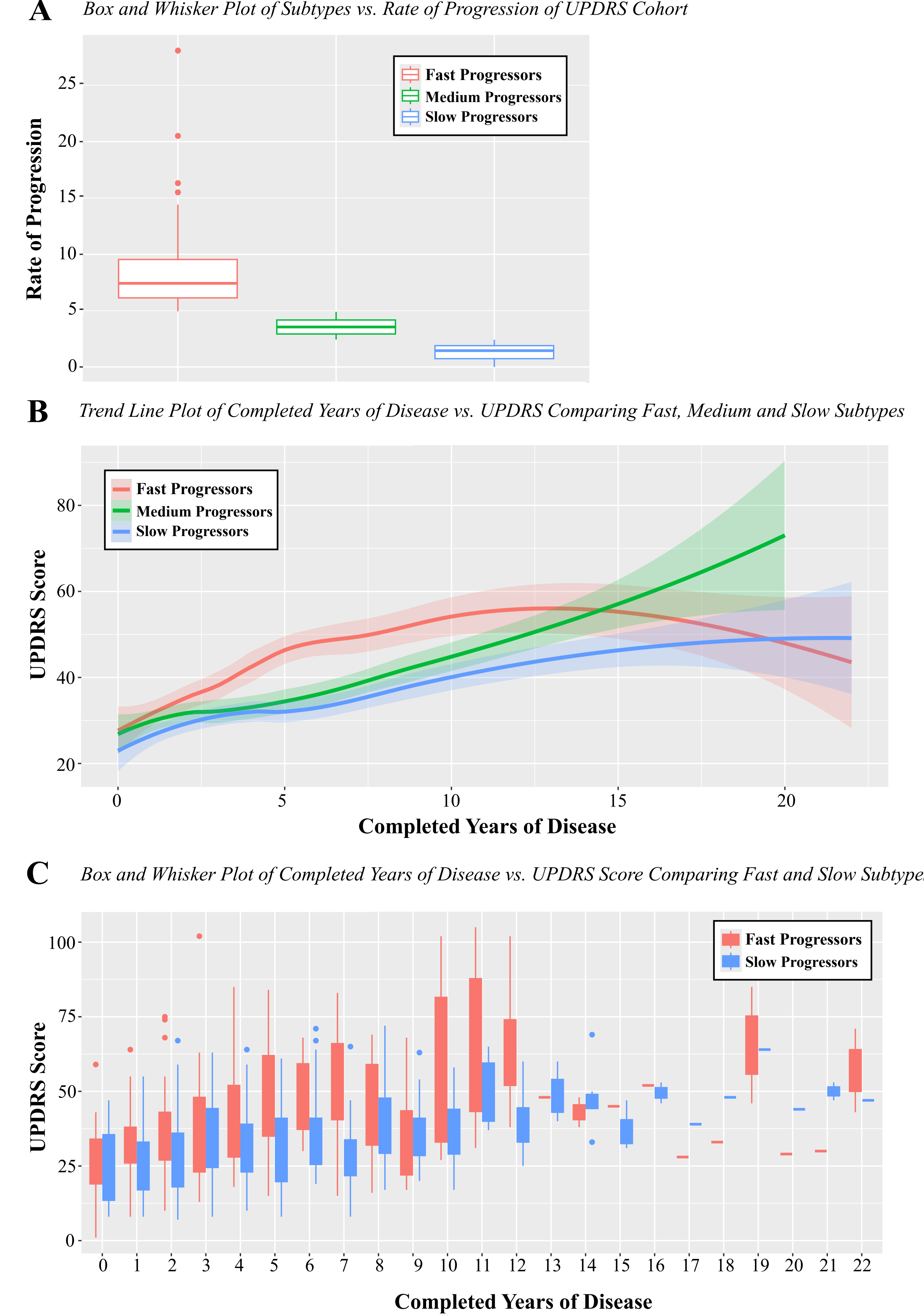
Severity Index Construction from a Pairwise Analysis



- The pairwise approach appeared to be a **novel statistical** method of highlighting the most important features in the dataset for assessing severity
 - By training a **ML model** to predict whether the comparisons between **2 timepoints** of patient data are either **forwards or backwards** in time, the most clinically significant features were uncovered
 - Then the **feature weights** of this model with the **highest accuracy** were used to construct an **overall severity index**

4. Results

Modeling of Patient Subtypes



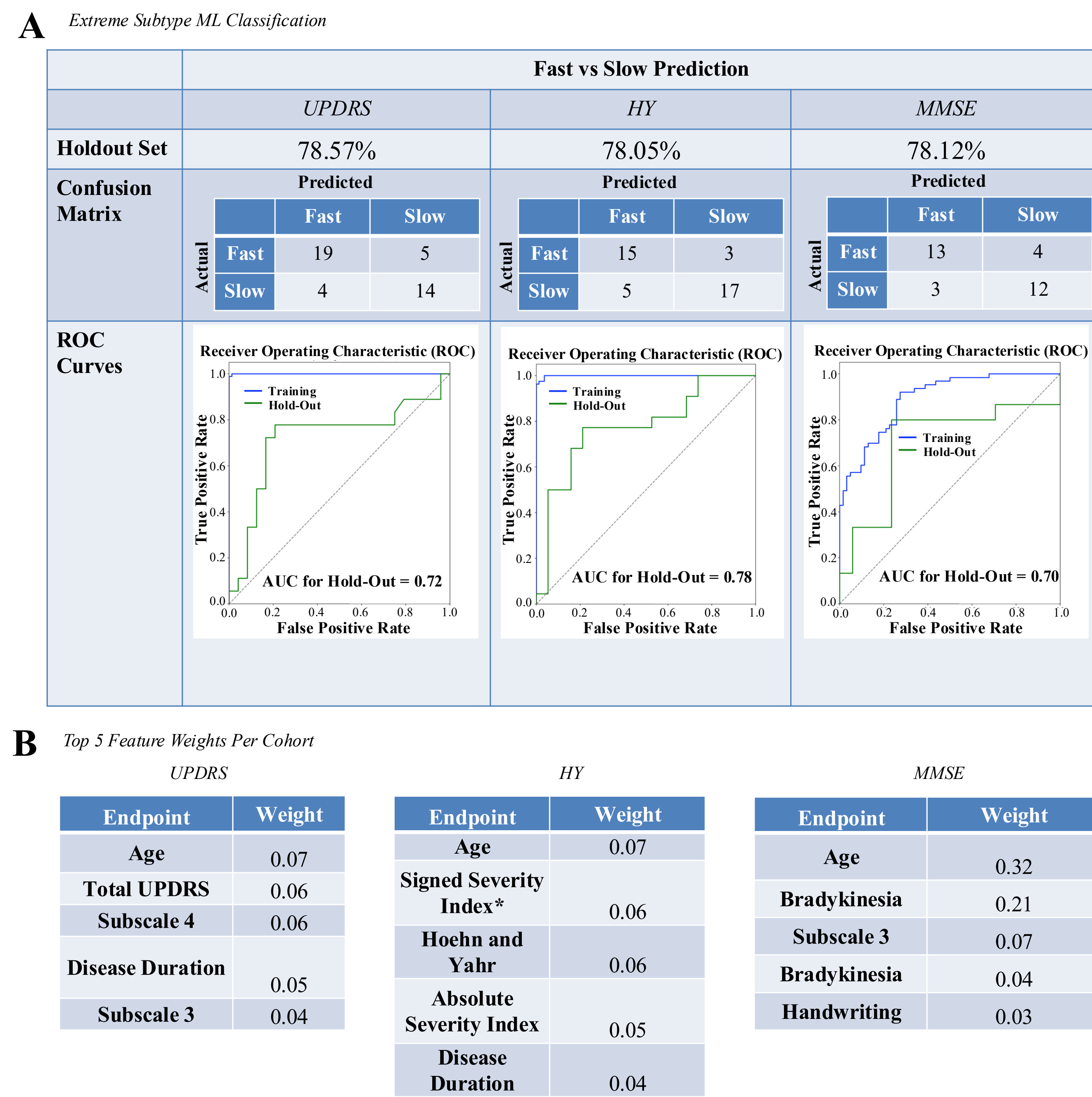
- *Note similar modeling was conducted for MMSE and HY progression cohort;
- Figure A shows the **different rates of progression (from slopes)** for each PD subtype and figures B and C highlight the **longitudinal distinction** between the subtypes
 - Significant differences were observed among all the groups **Kruskal-Wallis H Test and Dunn's test** revealing significant pairwise differences ($p < 0.001$)

Severity Index Construction from a Pairwise Analysis



- All three cohorts demonstrated high testing accuracy, exceeding 80%
- Dyskinesia, posture, and overall HY score** were identified as the most clinically significant predictors

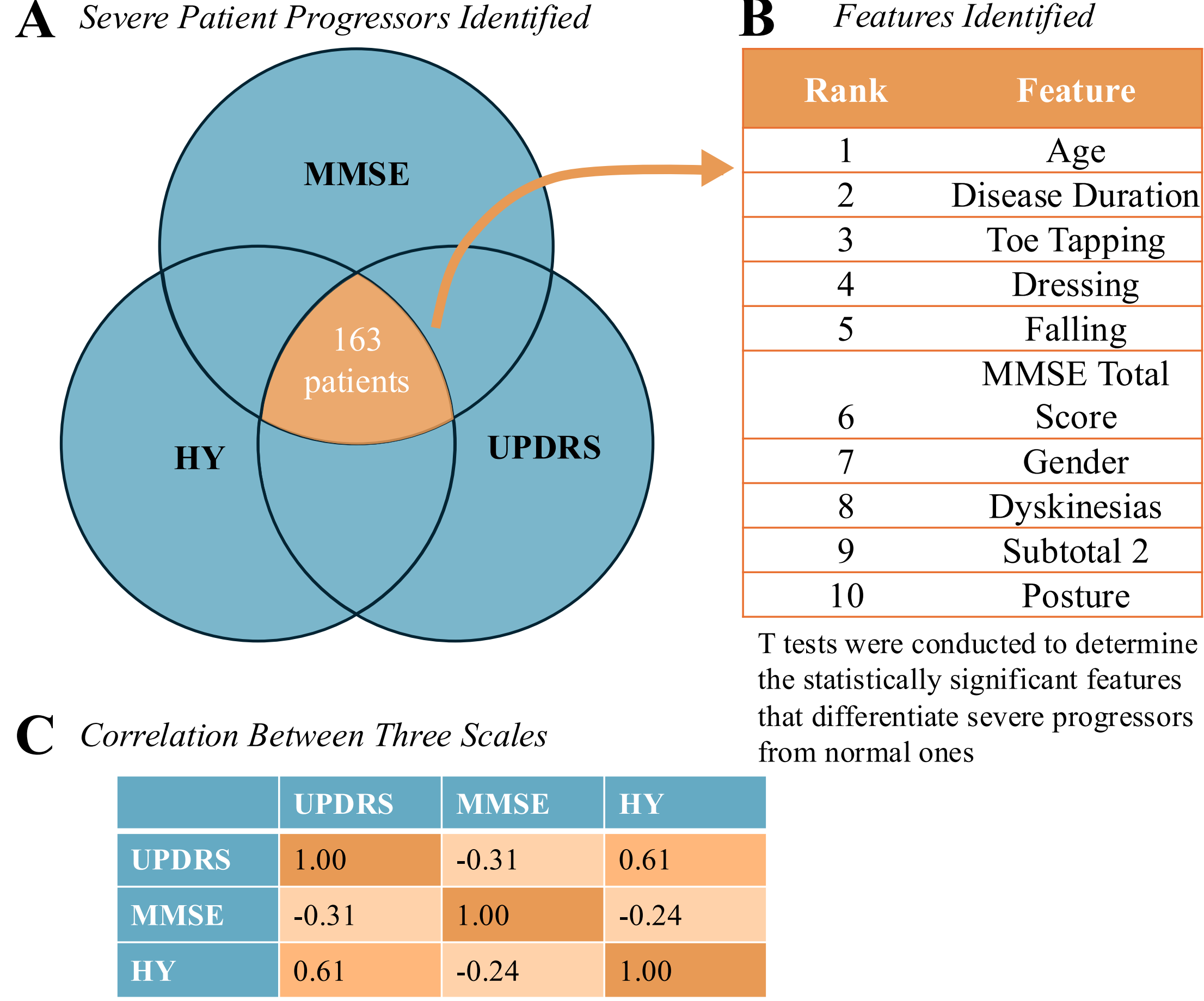
Predicting Based on Baseline Characteristics



*The summative severity index from pairwise analysis included positive and negative weights, while the absolute severity index used their absolute values, and the signed severity index used both positive and negative weights.

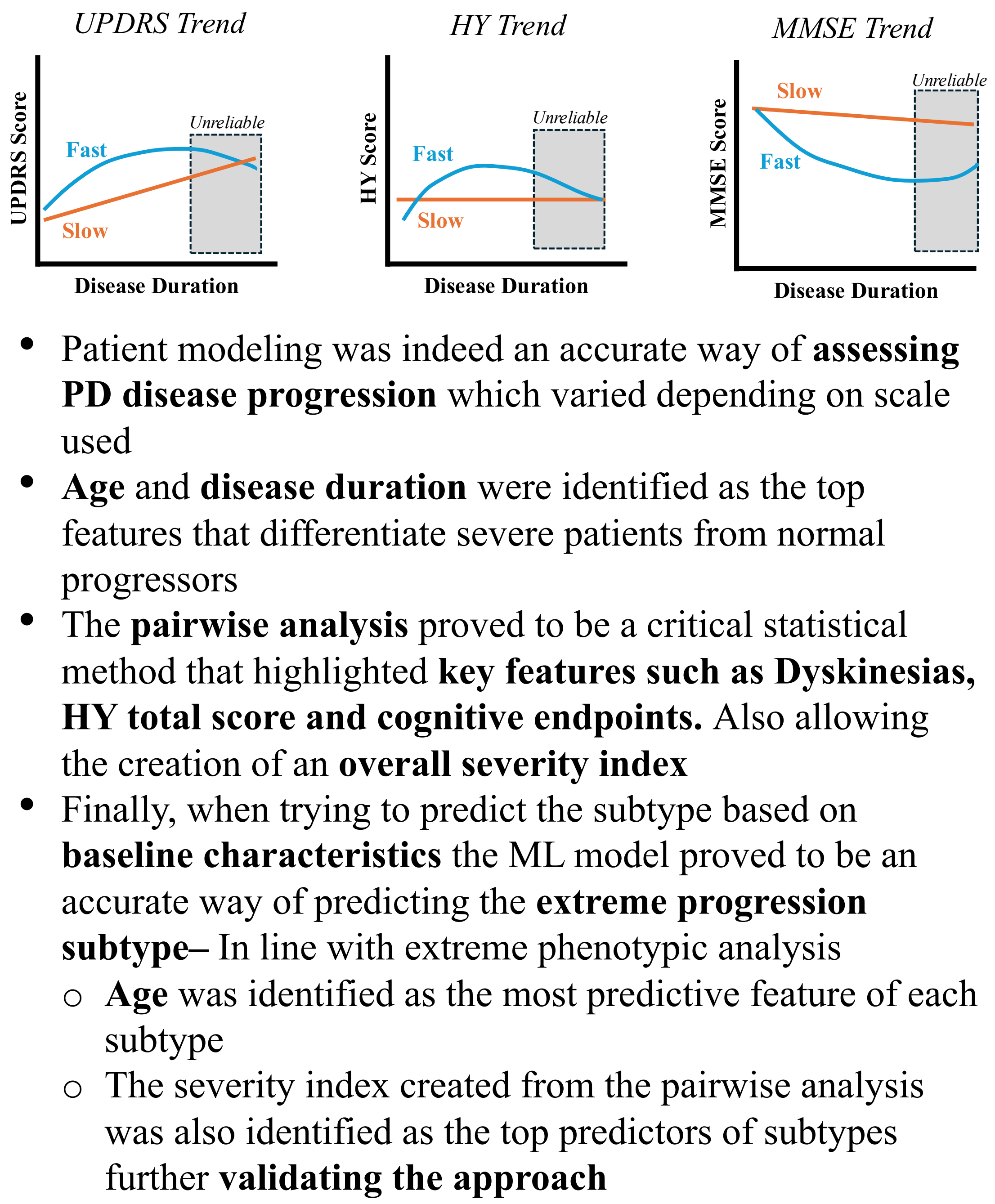
- High testing accuracy (~78%)** was achieved across cohorts
- Age and Severity Index derived from the Pairwise analysis were top features

Clinical Scale Relationship



- Severe patients identified who progressed by all three clinical scales and T tests performed to determine **clinical endpoints** that differentiate

5. Conclusion and Summary



6. Future Implications

- Personalized treatment approaches** as stratifying PD patients by progression subtype could enable tailored treatment plans
- Identifying key **clinical features** associated with progression subtypes can be combined with **multi-omics data** such as genomic, or proteomic to uncover novel biomarkers
- Refines diagnostic tools to **track disease progression** in a clinical setting

7. Next Steps

- The cohort sizes, particularly for **MMSE (n=236)** may limit the statistical power and **generalizability** of the findings
- Lack of demographic diversity as it was **predominantly white**
- Need for **external validation** in an independent, multi-center longitudinal cohort to confirm its reproducibility

8. References

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